

Technical Summary:

Tree of Thoughts: Deliberate Problem Solving with Large Language Models

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1 Research Problem and Motivation

Standard inference with Large Language Models (LLMs) is dominated by left-to-right decoding that produces a single reasoning trajectory. This decoding pattern is limited on tasks that require explicit exploration of alternatives, systematic backtracking, and lookahead-driven decisions. Existing prompting approaches that elicit intermediate text, such as Chain-of-Thought (CoT), still follow a linear path and therefore cannot directly represent branching search. Sampling many independent solutions improves coverage but does not maintain a persistent search space over intermediate decisions, which is central to classical problem solving.

The target setting is deliberate problem solving at inference time, where an LLM is used both to generate candidate intermediate steps and to guide a structured search procedure over intermediate reasoning states. The gap addressed is the absence of an explicit, modular search-and-evaluation loop for LLM reasoning that can adapt to task structure and resource constraints without additional training.

2 Related Work

2.1 Prompting baselines

Input-Output prompting (IO) directly maps input to output with instructions and in-context examples. Chain-of-Thought prompting (CoT) augments outputs with intermediate reasoning steps, but typically generates a single linear chain. Self-Consistency for CoT (CoT-SC) aggregates multiple independently sampled chains by majority vote, improving robustness but remaining a trajectory-level sampling method rather than a step-wise search method. Iterative refinement augments an initial solution with subsequent revisions conditioned on feedback; in the Game of 24 setup, the paper reports refinement using ground-truth correctness feedback.

2.2 Search and heuristic guidance

Classical search procedures such as Breadth-First Search (BFS) and Depth-First Search (DFS) support branching, pruning, and backtracking, but depend on heuristics for efficiency. The studied approach uses an LLM to supply both (i) candidate intermediate steps and (ii) heuristic evaluations of partial states, enabling search over intermediate natural-language reasoning units.

2.3 Differences introduced by the approach

The approach introduces (i) an explicit representation of intermediate reasoning units as thoughts, (ii) a modular interface for generating and evaluating thought candidates, and (iii) integration with explicit search

Table 1: Datasets and construction details.

Dataset / Task	Source and construction	Scale, split, and constraints
Game of 24	Scraped from <code>4nums.com</code> ; games ordered by human solving time from easy to hard.	1,362 total games; evaluation uses games indexed 901–1,000 (100 games). Each instance contains four numbers; success requires an equation that equals 24 and uses each input number exactly once with arithmetic operations.
Creative Writing	Inputs sampled from <code>randomwordgenerator.com</code> .	100 inputs. Input consists of four random sentences; output is a coherent passage with four paragraphs ending with the four input sentences, respectively. No ground-truth targets.
5×5 Mini Crosswords	Scraped from GooBix.	156 puzzles total; evaluation uses 20 puzzles with indices 1, 6, . . . , 96; prompting uses 5 puzzles with indices 136, 141, 146, 151, 156. Each task includes 10 clues (5 across, 5 down) and requires a 5×5 board (25 letters).
GSM8K	Not specified in the paper.	Evaluation on a subset of 100 random test questions; dataset construction and preprocessing are not specified in the paper.
StrategyQA	Not specified in the paper.	Evaluation on a subset of 100 random development questions; dataset construction and preprocessing are not specified in the paper.

algorithms (BFS and DFS) that retain and revisit intermediate states. The paper also interprets IO, CoT, CoT-SC, and refinement as special cases that arise by restricting tree depth, breadth, or revision operations.

3 Dataset Construction

The study evaluates inference-time procedures on three tasks designed to challenge GPT-4 under standard prompting and reports additional experiments on GSM8K and StrategyQA. Table 1 summarizes dataset sources and splits as specified. All missing fields are reported as *Not specified in the paper*.

4 Query Protocol and Task Definitions

4.1 General protocol

Each task is formatted as a text input x and a text output y . Prompts provide task instructions and, in some settings, in-context examples. The evaluated methods differ in how intermediate reasoning is generated,

Table 2: Task definitions and evaluation metrics.

Task	Input, output, and success definition	Metrics and meaning
Game of 24	Input: four numbers. Output: an arithmetic expression that equals 24 using each number exactly once. Success: a valid equation satisfying the constraint.	Success rate (%): fraction of cases solved; higher is better.
Creative Writing	Input: four random sentences. Output: a four-paragraph passage ending with the four input sentences, respectively. Success: coherence rather than exact match.	Coherency score (1–10): GPT-4 assigns a scalar score; five independent scores are averaged. Human pairwise comparison counts: number of inputs where CoT is preferred, outputs are similar, or the approach is preferred.
5×5 Mini Crosswords	Input: 10 clues (5 across, 5 down). Output: a 5×5 letter grid. Success: correct grid under clue constraints.	Letter (%): fraction of 25 letters correct. Word (%): fraction of 10 clue-words correct. Game (%): fraction of puzzles fully solved.

retained, and revised during inference.

4.2 Task definitions, success criteria, and metrics

Table 2 specifies the task interface and success standards. For tasks with exact targets (Game of 24 and Mini Crosswords), success is defined by correctness under task constraints. For open-ended generation (Creative Writing), success is defined by coherence measured through LLM scoring and human preferences.

4.3 Evaluation settings and reported hyperparameters

Unless otherwise stated, experiments use GPT-4 in Chat Completion mode with sampling temperature 0.7. The study evaluates inference-time procedures without additional training. Table 3 consolidates settings explicitly stated.

5 Modeling Approach

5.1 Tree of Thoughts framework

The approach structures inference as a search tree where each node corresponds to a partial reasoning state. A thought is a coherent intermediate unit, such as an intermediate equation, a short plan, or a partial crossword assignment. A state is represented as $s = [x, z_{1..i}]$, where x is the original input and $z_{1..i}$ are the first i thought steps.

The framework decomposes into four modules: (i) thought decomposition (choice of depth and granularity), (ii) thought generation (proposing multiple candidate next thoughts), (iii) state evaluation (assigning heuristic scores or votes), and (iv) search (BFS or DFS) that uses evaluation feedback to guide exploration.

Table 3: Inference-time settings. Missing fields are reported as *Not specified in the paper*.

Item	Reported setting
Base model and decoding	GPT-4 (Chat Completion mode); temperature 0.7.
Game of 24 (approach)	Thought depth 3 (intermediate equations). BFS with breadth $b = 5$ (also reported $b = 1$). Value evaluator returns labels <code>sure/maybe/impossible</code> ; value sampling repeated 3 times per candidate thought.
Creative Writing (approach)	Two-stage sample-and-vote: sample 5 plans and vote 5 times to select a plan, then sample 5 passages conditioned on the selected plan and vote 5 times to select an output.
Mini Crosswords (approach)	DFS with pruning and backtracking; thoughts constrained not to change filled words or letters; at most 10 intermediate steps; DFS limited to 100 search steps.
Baselines	IO and CoT prompts with 5-shot examples. CoT-SC uses $k = 100$ samples. Game of 24 includes IO+Refine with up to 10 iterations using ground-truth correctness feedback; “best of 100” oracle-style evaluation is reported for IO and CoT.

5.2 Core equations (core-only, purpose-driven)

Thought generation.

$$z_{i+1}^{(j)} \sim p_{\theta}^{\text{CoT}}(z_{i+1} \mid s) = p_{\theta}^{\text{CoT}}(z_{i+1} \mid x, z_{1\dots i}). \quad (1)$$

Purpose. Generates multiple candidate next thoughts conditioned on the current state, enabling branching at each step.

Symbols. x : task input; $z_{1\dots i}$: previously generated thoughts; $s = [x, z_{1\dots i}]$: current state; $z_{i+1}^{(j)}$: the j -th candidate next thought; $p_{\theta}^{\text{CoT}}(\cdot)$: LLM distribution induced by a CoT-style prompt.

Intuition. Multiple plausible next steps are proposed so that search can explore alternatives rather than committing early.

Usage. Used at each search node to create candidate child states; in some tasks, a dedicated proposal prompt is used to improve candidate diversity and quality.

State evaluation (value interface).

$$V(p_{\theta}, \{s\})(s) \sim p_{\theta}^{\text{value}}(v \mid s). \quad (2)$$

Purpose. Produces heuristic scores or labels that rank and prune partial states during search.

Symbols. s : a state; v : a value label or score; $V(p_{\theta}, \{s\})(s)$: evaluator output for state s ; $p_{\theta}^{\text{value}}(v \mid s)$: LLM distribution induced by a value-evaluation prompt.

Intuition. The LLM acts as a heuristic judge estimating whether a partial solution is extendable to a complete solution.

Usage. In Game of 24, the value prompt yields `sure/maybe/impossible` and is sampled multiple times per candidate; in Mini Crosswords, evaluation supports pruning when remaining clues are judged impossible under current letter constraints.

Table 4: Game of 24 results (success rate).

Method	Success (%)
IO prompt	7.3
CoT prompt	4.0
CoT-SC ($k = 100$)	9.0
ToT (BFS, $b = 1$)	45
ToT (BFS, $b = 5$)	74
IO + Refine ($k = 10$)	27
IO (best of 100)	33
CoT (best of 100)	49

BFS selection rule.

$$S_{t+1} = \arg \max_{S \subseteq S'_t} \sum_{s \in S} V(p_\theta, S_t)(s) \quad \text{s.t.} \quad |S| = b. \quad (3)$$

Purpose. Selects a breadth-limited frontier of size b after expansion, implementing deliberate search with bounded computation.

Symbols. t : search step; S_t : current frontier; S'_t : expanded candidates; b : breadth; $V(p_\theta, S_t)(s)$: evaluation score of s .

Intuition. The procedure retains a small set of promising partial solutions rather than exploring all branches.

Usage. Used for shallow tasks such as Game of 24 and for planning-selection in Creative Writing.

5.3 Minimal end-to-end example

For an instance with numbers 4 9 10 13, one valid solution is $(10-4) * (13-9) = 24$. With thought depth 3, an example trajectory is: $13-9=4$ (left: 4 4 10) $\rightarrow 10-4=6$ (left: 4 6) $\rightarrow 4*6=24$ (left: 24). At each step, multiple candidate intermediate equations are proposed, evaluated for promise, and selected via breadth-limited BFS.

6 Empirical Results

6.1 Main results on three tasks

Table 4 reports success rates on Game of 24. Standard baselines are low (IO 7.3%, CoT 4.0%, CoT-SC 9.0%). The approach improves performance with explicit search (45% at $b = 1$ and 74% at $b = 5$) and exceeds an oracle-style “best of 100” CoT baseline (49%). The paper reports that around 60% of CoT samples fail after producing the first step, indicating that early-step errors dominate linear reasoning failures.

Creative Writing is evaluated without ground-truth targets. The paper reports GPT-4 coherency scores and a blind human preference study comparing CoT and the approach. Reported coherency scores are 6.19 (IO), 6.93 (CoT), and 7.56 (approach). With refinement, scores are 7.67 (IO) and 7.91 (approach). Human comparisons over 100 inputs yield 41 cases where CoT is preferred, 21 cases judged similar, and 38 cases where the approach is preferred.

Table 6 reports Mini Crosswords results. IO and CoT achieve low word-level accuracy (14.0% and 15.6%), while the approach reaches 60.0% word accuracy and solves 20% of puzzles. An oracle variant that outputs the best explored DFS state increases the game solve rate to 35%. Ablations show that removing pruning or backtracking degrades performance, with a large drop under removal of backtracking.

Table 5: Creative Writing results as reported.

Metric	Result
IO coherency score (baseline)	6.19
CoT coherency score (baseline)	6.93
ToT coherency score (baseline)	7.56
IO coherency score (+ refine)	7.67
ToT coherency score (+ refine)	7.91
Human comparison (CoT preferred / Similar / ToT preferred)	41 / 21 / 38

Table 6: 5×5 Mini Crosswords results (percentages).

Method	Letter	Word	Game
IO	38.7	14.0	0
CoT	40.6	15.6	1
ToT	78.0	60.0	20
ToT + best state	82.4	67.5	35
ToT - prune	65.4	41.5	5
ToT - backtrack	54.6	20.0	5

6.2 Additional experiments and model transfer

Table 7 summarizes additional results reported for GSM8K and StrategyQA under a simple zero-shot extension, and for GPT-3.5 on the invented tasks. The paper further reports mixed-model results on Game of 24 showing that generation quality is a bottleneck: GPT-4 generation with GPT-3.5 evaluation attains 64% success, while GPT-3.5 generation with GPT-4 evaluation attains 31%.

6.3 Cost and efficiency

The approach increases computation relative to IO and CoT. Table 8 reports token usage and cost per case for Game of 24 and Creative Writing. On Game of 24, the approach costs \$0.74 per case while achieving 74% success, whereas best-of-100 CoT costs \$0.47 per case while achieving 49% success. On Creative Writing, the approach costs \$0.32 per case versus \$0.06–\$0.07 for IO and CoT, reflecting multi-branch generation and repeated evaluation.

7 Summary

The study evaluates an inference-time framework that introduces explicit search over intermediate reasoning units generated by an LLM and guided by LLM-based evaluations. Across three challenging tasks, explicit search yields large gains over linear prompting baselines: 74% success on Game of 24 with breadth-limited BFS compared to 7.3% (IO), 4.0% (CoT), and 9.0% (CoT-SC); 60.0% word accuracy and 20% full-solve rate on Mini Crosswords compared to 14.0% and 15.6% word accuracy for IO and CoT; and improved coherency outcomes on Creative Writing under both automatic scoring and human comparison.

The limitations explicitly evidenced include increased computational overhead (higher token usage and cost), dependence on thought generation quality (generation bottleneck under mixed-model experiments on

Table 7: Additional results reported.

Setting	Result
GSM8K (GPT-4)	Accuracy (%): IO 51, CoT 86, ToT 90.
StrategyQA (GPT-4)	Accuracy (%): IO 73, CoT 82, ToT 83.
Game of 24 (GPT-3.5)	Success (%): IO 6, CoT 3, ToT 19.
Creative Writing (GPT-3.5)	Coherency score: IO 4.47, CoT 5.16, ToT 6.62.
Game of 24 (mixed models)	Success (%): GPT-4 generation + GPT-3.5 evaluation 64; GPT-3.5 generation + GPT-4 evaluation 31.

Table 8: Cost analysis reported (tokens and cost per case).

Task	Method	Tokens (gen/prompt)	Cost and performance
Game of 24	IO (best of 100)	1.8k / 1.0k	\$0.13 per case; 33% success.
Game of 24	CoT (best of 100)	6.7k / 2.2k	\$0.47 per case; 49% success.
Game of 24	ToT	5.5k / 1.4k	\$0.74 per case; 74% success.
Creative Writing	IO	0.9k / 0.4k	\$0.06 per case; performance not specified in the paper.
Creative Writing	CoT	0.9k / 0.4k	\$0.07 per case; performance not specified in the paper.
Creative Writing	ToT	4.0k / 2.9k	\$0.32 per case; performance not specified in the paper.

Game of 24), and imperfect evaluation heuristics that can prune solvable states in Mini Crosswords, particularly in the presence of rare or obsolete words as stated by the paper. The oracle best-state results indicate headroom in output selection heuristics, and ablations confirm that pruning and backtracking materially affect performance in deep search settings.